

Emergent Suspiciousness in Trained Monitors

Prompt Injection Shortcuts,
Context-Dependent Robustness, and Data Augmentation

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Our approach: distill the monitoring decision directly

We reuse Sinha et al.'s trajectory data framework: 8,688 training examples.

Sinha et al.: SFT + RL

Learn to generate a judgment

SFT on a monitor rationale and a final 0–10 score.

GRPO rewards calibration and ranking against binary labels.

Output: a rationale and score.

Ours: logit distillation

Learn the teacher's decision probability

Train a Qwen LoRA adapter with soft binary cross-entropy.

Supervise the 0/1 decision logits directly.

Output: a harmfulness probability.

Our adapters use soft decision targets, without rationale supervision or RL.

Sinha et al. (2026), Appendix D. Training provenance and input caveats are documented in `sources.md`.

From teacher probabilities to a small monitor

Teacher: Kimi K3

Read the trajectory under the full monitoring rubric.

Normalize the 0/1 logprobs to obtain a soft harmfulness target q .

Student: Qwen3.5-4B + LoRA

Read the same trajectory under a compact instruction.

Convert decision logits to a score:
 $p_\theta = \sigma(z_1 - z_0)$.

Train the adapter to match the teacher's probability:

$$\mathcal{L} = -q \log p_\theta - (1 - q) \log(1 - p_\theta)$$

Baseline: 8,688 trajectories, one epoch; frozen backbone. At inference, read the decision logits directly.

Our question: does the monitor learn harmful behavior, or become suspicious of prompt-injection text itself?